

S-CARD-CMSA: Participant Information and Method Description

IEEE CEC 2026 Competition on Benchmarking Niching Methods for Multimodal Optimization

Method report for final result submission

1. Participant Information

Participant	Affiliation	Country/Region	Contact
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2. Overview of the Used Method

The submitted method is S-CARD-CMSA, which denotes a Score-aware Candidate Archive with Density-filtered Reporting framework built on RS-CMSA-ESII. RS-CMSA-ESII is used as the base multimodal optimizer. The proposed method does not redesign the core optimizer; instead, it improves candidate retention and final reporting.

The core RS-CMSA-ESII search components are kept unchanged, including sampling, covariance matrix adaptation, mutation-strength adaptation, taboo-region handling, restart logic, termination criteria, bound handling, and hill-valley-based archive checking. The modification is applied after and around restarts through a passive secondary candidate archive and a final score-aware density-filtered reporting rule.

True global-minimum locations are not used during optimization, candidate selection, restart initialization, archive update, or final reporting. They were used only in offline development analysis for computing RPR, precision, F1-score, and score.

3. Main Algorithmic Components

Component	Description
Base optimizer	RS-CMSA-ESII is used as the search engine. Each restart evolves one covariance-adaptive subpopulation and updates the primary archive using the original archive and hill-valley checking logic.
Primary archive	The original RS-CMSA-ESII primary archive is preserved. Primary-archive candidates are treated as high-confidence candidates during final reporting.
Passive secondary candidate archive	At the end of each restart, the restart-best candidate and its objective value are stored in a secondary archive. This archive is passive: it does not define taboo regions, modify sampling, change covariance adaptation, or affect restart decisions.
Candidate pool	Before final reporting, invalid or non-finite secondary candidates are removed. The final candidate pool is formed from the union of the primary archive and the feasible secondary archive.
Density-filtered reporting	Primary archive candidates are inserted first. Secondary candidates are then considered in ascending objective value and are retained only if sufficiently separated from the current reported set. If a secondary candidate is close to an existing reported candidate, the better objective-value representative is retained without removing protected primary candidates.

4. Final Reporting Rule

Let A denote the primary archive and C_f denote the feasible secondary archive. The final candidate pool is $P = A \cup C_f$. Candidate similarity is measured using normalized Euclidean distance,

$$d_{norm}(x_i, x_j) = ||(x_i - x_j) / (u - l)||_2$$

$$\rho(x) = \min_{\{y \in R\}} d_{norm}(x, y)$$

$$\text{insert } x \text{ into } R \text{ if } \rho(x) \geq \tau_\rho(D)$$

$$\tau_\rho(D) = \alpha_\rho \sqrt{D}$$

Here, u and l are the upper and lower bounds, R is the current reported set, $\tau_\rho(D)$ is the dimension-scaled density threshold, and α_ρ is the fixed density-filtering coefficient. The selected setting uses $\Delta_f = 2e - 3$ and $\alpha_\rho = 2e - 3$ for all problems, instances, and dimensions.

5. Parameter Setting

Parameter / Setting	Value	Purpose / Comment
Benchmark cases	PID = 1,...,16; PIN = 1,...,15; D in {2, 5, 10, 20}	Official CEC 2026 setup
Function-evaluation budget	FE_max = 20,000 D	Official competition budget
Base optimizer	RS-CMSA-ESII	Original search engine retained
Core RS-CMSA-ESII parameters	Unchanged from the base implementation	No retuning of sampling, covariance adaptation, taboo update, restart logic, or termination
Secondary archive	Enabled	Stores restart-best candidate and objective value from each restart
Secondary archive influence	Passive only	Does not influence search trajectory or taboo regions
Candidate value window	$\Delta_f = 2e - 3$	Filters clearly inferior secondary candidates
Density coefficient	$\alpha_\rho = 2e - 3$	Selected balanced density-filtering coefficient
Density threshold	$\tau_\rho(D) = \alpha_\rho \sqrt{D}$	Dimension-scaled duplicate-control threshold
Distance metric	Normalized Euclidean distance	Uses search-space bounds for scale normalization
Primary archive protection	Enabled	Secondary candidates do not remove protected primary candidates
Final selected reporting rule	DF-SCA-Balanced	Density-filtered score-aware reporting with $\alpha_\rho = 2e-3$
Use of true global minimizers	Not used during optimization or reporting	Used only for offline analysis and validation

6. Density-Filtering Sensitivity

Three density-filtering coefficients were evaluated on the 320-run development subset. The balanced setting was selected because it provided the best mean score while preserving RPR and improving precision and F1-score.

Variant name	alpha_rho	Mean RPR	Mean precision	Mean F1	Mean score	Mean N_sol
DF-SCA-Low	0.001	0.558904	0.849994	0.643001	0.600953	10.153
DF-SCA-Balanced (selected)	0.002	0.558904	0.870491	0.650922	0.604913	9.819
DF-SCA-High	0.005	0.558954	0.810685	0.637740	0.598347	10.494

7. Development and Validation Summary

Experiment stage	Cases	Role
320-run development subset	PID in {1,2,6,7,10,11,14,15}, PIN in {1,5,10,15}, D in {10,20}, runs 1,...,5	Used for reporting-rule comparison and density-coefficient selection
768-run broader validation subset	PID = 1,...,16, PIN in {1,5,10,15}, D in {2,5,10,20}, runs 1,2,3	Used to verify that the selected DF-SCA rule remains stable across additional PIDs and dimensions
Complete submission scale	16 x 15 x 4 = 960 PID-PIN-dimension combinations	Used to generate the final competition output files

8. Submission and Reproducibility Notes

- The final submitted method is S-CARD-CMSA with DF-SCA-Balanced reporting.
- The final output consists of one CSV file for each official PID-PIN-dimension case, resulting in 960 files.
- Each CSV file reports decision vectors and the corresponding objective value. The number of decision-variable columns equals D, and the final column is the objective value.
- The method uses the official objective-function evaluation budget $FE_max = 20,000D$.
- The proposed additions do not consume extra objective-function evaluations for search; they use candidate information already generated by RS-CMSA-ESII.
- The method is intended as a conservative and reproducible archive/reporting-level enhancement of the RS-CMSA-ESII optimizer.

References

- [1] A. Ahrari, K. Deb, and M. Preuss, "Multimodal optimization by covariance matrix self-adaptation evolution strategy with repelling subpopulations," *Evolutionary Computation*, vol. 25, no. 3, pp. 439-471, 2017.
- [2] A. Ahrari, S. Elsayed, R. Sarker, D. Essam, and C. A. Coello Coello, "Static and dynamic multimodal optimization by improved covariance matrix self-adaptation evolution strategy with repelling subpopulations," *IEEE Transactions on Evolutionary Computation*, vol. 26, no. 3, pp. 527-541, 2022.
- [3] A. Ahrari, J. E. Fieldsend, M. Preuss, X. Li, and M. G. Epitropakis, "Experimental setup for CEC 2026 competition on benchmarking niching methods for multimodal optimization," *Tech. Rep. CIS-TF-MMO-TR-2026-001*, IEEE CIS Task Force on Multimodal Optimization, 2026.